

Research Methodology

B.Sc (Hons) Mathematics – Semester VI

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6.1 Audience

- Audience refers to the **group of readers** for whom the paper is written.
- Understanding the audience helps adjust **language, tone, and level of explanation**.
- Two main types:
 - **Primary audience:** main readers such as students or researchers.
 - **Secondary audience:** others interested in the topic.
- Knowing the audience helps **engage readers and communicate ideas effectively**.



6.2 Organization and Structure

- Organization means arranging ideas in a **logical and clear order**.
- A good structure improves **clarity and readability**.
- Common structure includes:
 - **Introduction**
 - **Body**
 - **Conclusion**

- Types of organization:
- Chronological order
- Spatial order
- Order of importance
- Compare and contrast

6.3 Title

- The title is the **first thing readers see** in a research paper.
- It should clearly reflect the **main topic or idea** of the paper.
- A good title should be:
 - Relevant**
 - Concise**
 - Interesting**
- Titles help readers quickly understand **what the research is about**.



6.4 Author List

- The author list identifies **who wrote the research paper**.
- It gives proper **credit to contributors**.
- Authors may be listed **alphabetically or based on contribution**.
- Author affiliations such as **universities or institutions** may also be included.
- Proper author listing improves **credibility and transparency**.

6.5 Date

- Including the date helps readers know **when the work was written**.
- It helps distinguish **different versions or drafts** of the paper.
- The date is usually written **after the author's name**.
- Writing the **month in words** avoids confusion between date formats.



6.6 Abstract

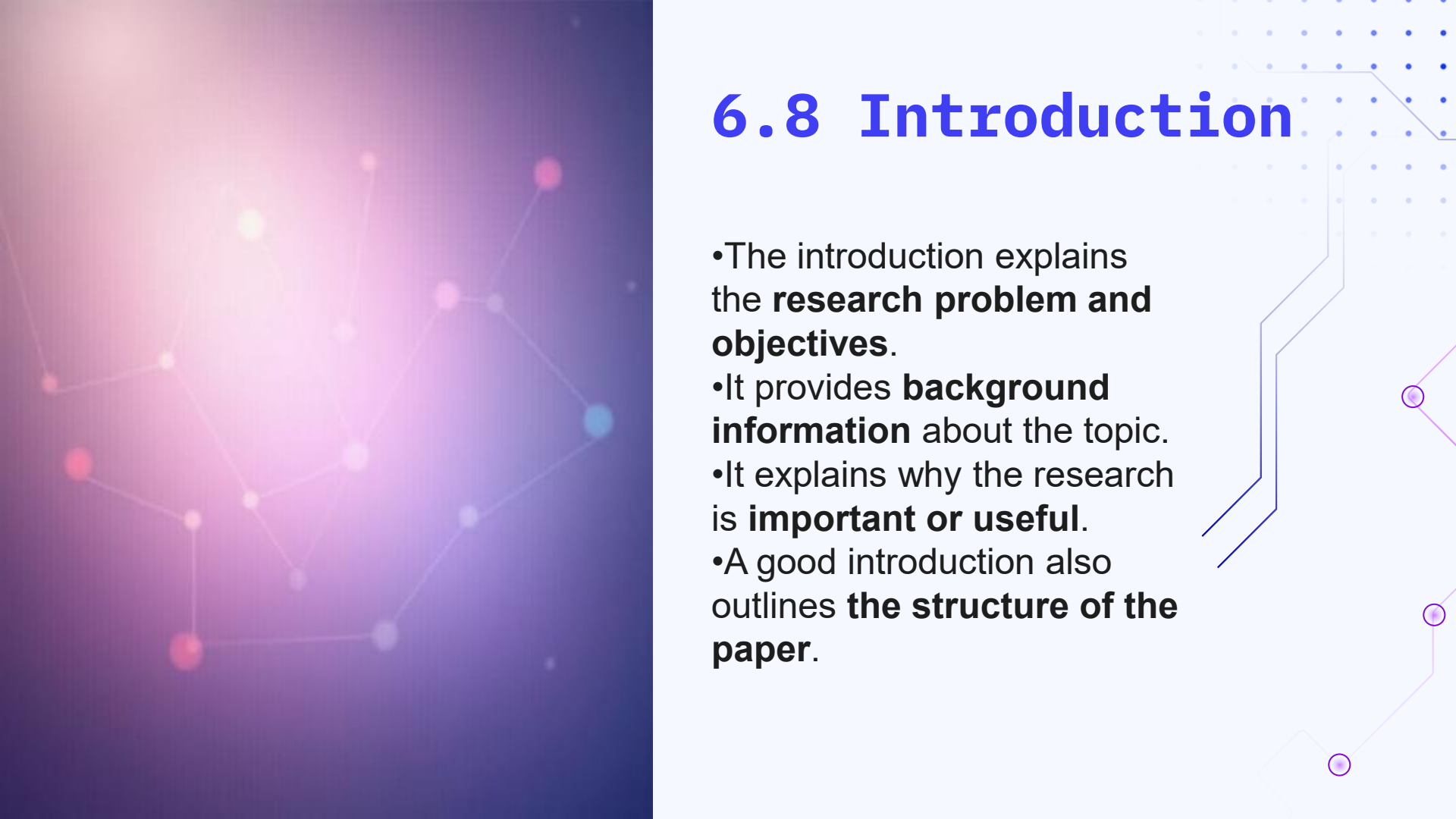
- The abstract is a **brief summary of the research paper.**
- Usually **200–300 words long.**
- It helps readers decide **whether to read the full paper.**

It describes:

- The research problem
- Methods used
- Main results

6.7 Keywords and Subject Classification

- Keywords are **important words describing the research topic**.
- They help search engines and databases **find the paper easily**.
- Usually **5–10 keywords** are included.
- Subject classification systems categorize research into **specific academic fields**.



6.8 Introduction

- The introduction explains the **research problem and objectives**.
- It provides **background information** about the topic.
- It explains why the research is **important or useful**.
- A good introduction also outlines **the structure of the paper**.

6.9 Computational Experiments

- Computational experiments are used to **test algorithms and research methods**.
- They help verify **theoretical results**.
- Experiments should be **carefully designed and clearly explained**.
- Other researchers should be able to **repeat the experiments**.

6.10 Tables

- Tables present **numerical data in a clear format**.
- They allow readers to **compare values easily**.
- Tables should be **simple and well organized**.
- Each table should include a **clear caption or title**.



6.11 Citations

- Citations acknowledge **previous research and sources**.
- They help avoid **plagiarism**.
- Common citation styles include:
 - Numbered citation system
 - Author–date system
- Proper citations increase the **credibility of research**.

6.12 Conclusions

- The conclusion summarizes the **main findings of the research**.
- It explains the **importance of the results**.
- It may include **limitations of the study**.
- Suggestions for **future research** can also be included.





6.13 Acknowledgements

- This section thanks people who **helped or supported the research**.
- Funding agencies and organizations should be **properly acknowledged**.
- Advisors or reviewers may also be mentioned.
- It shows **gratitude and transparency in research**.

6.14 Appendix

- The appendix contains **additional supporting information**.
- It keeps the main paper **clear and concise**.

It may include:

- Detailed proofs
- Extra data
- Computer programs

6.15 Reference List

- The reference list includes **all sources cited in the paper.**
- It helps readers locate the **original sources.**
- References must follow **proper formatting rules.**
- Accurate references improve the **quality of research writing.**



6.16 Specifics and Academic Writing Rules

- Follow correct **capitalization and grammar rules**.
- Avoid **dangling theorems or unclear statements**.
- Use footnotes only when **necessary**.
- Number equations only when they are **referenced later**.
- Avoid **plagiarism and self-plagiarism** by properly citing sources.